

7B – Lecture 001 – Midterm 2 – Question 1

Rubric and Answer Key

Spring 2025

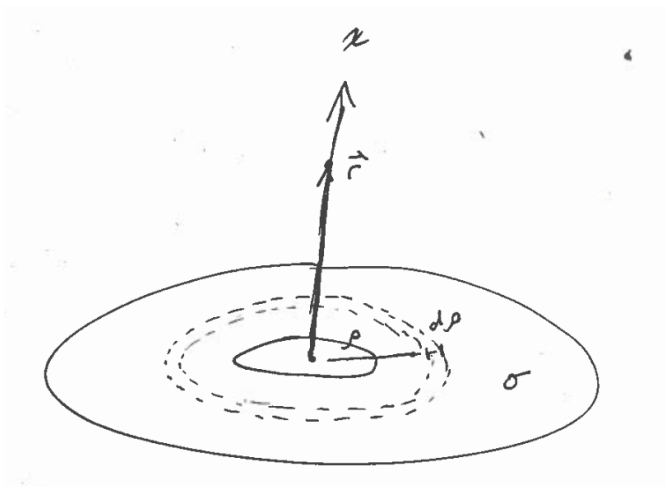
Problem 1 - 25 Points

This problem is split into four sections, (a), (b), (c), and (d), which I further split into sub-parts for the purpose of partial credit.

- (a) (11pts) Determine the electric potential produced by the annulus at any point on the x -axis.

Remark. This part can be done by doing a double integral over $\rho d\rho d\theta$ or by using the hint and splitting the annulus into thin rings. These are really the same thing – the rubric will go over the method using the hint, but is also applicable for the double integral method by performing the $d\theta$ integral first.

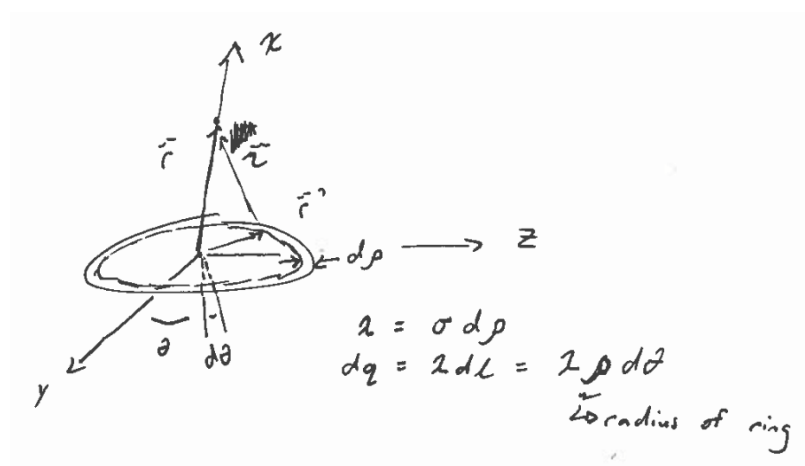
Sub-part 1. (2pts) split annulus into rings.



$$V = \int_{R_1}^{R_2} \frac{dV_{ring}(\rho)}{d\rho} d\rho$$

(or some equivalent expression) where $dV_{ring}(\rho)$ is the electric potential due to an infinitesimally ($d\rho$) thick ring of radius ρ and linear charge density $\lambda = \sigma d\rho$.

Sub-part 2. (4pts) electric potential of a ring.



The infinitesimal contribution of a point on the ring with charge dq to the electric potential dV_{ring} is

$$d(dV_{ring}) = \frac{1}{4\pi\epsilon_0} \frac{dq}{|\vec{z}|}$$

where $\vec{z} = \vec{r} - \vec{r}'$ is the displacement vector (the vector that points from the source to the point we are evaluating the potential). Here,

$$\vec{r} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}$$

is the point we are evaluating the electric potential, and

$$\vec{r}' = \begin{pmatrix} 0 \\ \rho \cos \theta \\ \rho \sin \theta \end{pmatrix}$$

is a point on the ring parameterized by θ . So, the magnitude

$$|\vec{z}| = \sqrt{\vec{z} \cdot \vec{z}} = \sqrt{x^2 + \rho^2 \cos^2 \theta + \rho^2 \sin^2 \theta} = \sqrt{x^2 + \rho^2}$$

Using this parameterization, the charge dq contained in an infinitesimal length $dl = \rho d\theta$ is

$$dq = \lambda dl = \lambda \rho d\theta$$

So, we can integrate

$$dV_{ring} = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\lambda \rho d\theta}{\sqrt{x^2 + \rho^2}} = \frac{1}{2\epsilon_0} \frac{\lambda \rho}{\sqrt{x^2 + \rho^2}}$$

Sub-part 3. (5pts) electric potential of annulus.

As the infinitesimal ring of thickness $d\rho$ has linear charge density $\lambda = \sigma d\rho$, we know

$$dV_{ring} = \frac{1}{2\epsilon_0} \frac{\sigma \rho d\rho}{\sqrt{x^2 + \rho^2}}$$

so

$$V = \frac{1}{2\epsilon_0} \int_{R_1}^{R_2} \frac{\sigma \rho}{\sqrt{x^2 + \rho^2}} d\rho = \frac{1}{4\epsilon_0} \int_{x^2 + R_1^2}^{x^2 + R_2^2} \frac{\sigma}{\sqrt{u}} du = \frac{\sigma}{2\epsilon_0} \left[\sqrt{x^2 + R_2^2} - \sqrt{x^2 + R_1^2} \right]$$

(b) (5pts) Determine the direction of the electric field on axis and using part (a), determine its magnitude at any point on the symmetry axis.

Sub-part 1. (2pts) direction.

By rotational symmetry, we know on the x axis the electric field must point in the $\pm \hat{x}$ direction. Assuming $\sigma > 0$, the electric field will be in the $+\hat{x}$ direction when $x > 0$ (this sentence not required).

Sub-part 2. (3pts) and magnitude.

$$\vec{E} = -\nabla V = -\hat{x} \frac{\partial}{\partial x} V - \hat{y} \frac{\partial}{\partial y} V - \hat{z} \frac{\partial}{\partial z} V$$

By symmetry, we know only the $\hat{x}$ component exists, so

$$\vec{E} = -\hat{x} \frac{\partial}{\partial x} V = \frac{x\sigma}{2\epsilon_0} \left[\frac{1}{\sqrt{x^2 + R_1^2}} - \frac{1}{\sqrt{x^2 + R_2^2}} \right] \hat{x}$$

(c) (7pts) Show that the electron undergoes simple harmonic motion about the center of the annulus.

Sub-part 1. (4pts) equation of motion.

$$F_x = m\ddot{x} = qE_x$$

$$m \frac{d^2x}{dt^2} = -\frac{e\sigma}{2\epsilon_0} \left[\frac{1}{\sqrt{x^2 + R_1^2}} - \frac{1}{\sqrt{x^2 + R_2^2}} \right] x$$

$$\frac{d^2x}{dt^2} + \frac{e\sigma}{2\epsilon_0 m} \left[\frac{1}{\sqrt{x^2 + R_1^2}} - \frac{1}{\sqrt{x^2 + R_2^2}} \right] x = 0$$

Sub-part 2. (1pts) small distance approximation.

As x is small compared to R_1 , we expand the equation of motion in x . Taylor expand

$$\frac{1}{\sqrt{x^2 + R^2}} = \frac{1}{R} \left(\frac{x^2}{R^2} + 1 \right)^{-1/2} \approx \frac{1}{R} \left(1 - \frac{1}{2} \frac{x^2}{R^2} + \dots \right)$$

Sub-part 3. (2pts) final equation of motion.

Keep only terms first order in x/R .

$$\frac{d^2x}{dt^2} + \frac{e\sigma}{2\epsilon_0 m} \left[\frac{1}{R_1} - \frac{1}{R_2} \right] x = 0$$

which is in the form of a simple harmonic oscillator

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

as $R_1 < R_2$ so $\left[\frac{1}{R_1} - \frac{1}{R_2} \right] > 0$.

(d) (2pts) Based on the equation obtained in part (c), determine the angular frequency ω of the oscillatory motion.

Sub-part 1. (2pts) take the square root of the coefficient of x in the equations of motion.

$$\omega = \sqrt{\frac{e\sigma}{2\epsilon_0 m} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]}$$

Problem 2 : rubric breakdown

a) (7 points)

2 pt: stating that total charge = $+q + \text{total negative charge} = 0$

2 pt: having total negative charge = $\int_a^{+\infty} \epsilon n(r) 4\pi r^2 dr$

1.5 pt: using the formula $\int x e^{-bx} dx = -\frac{e^{-bx}(bx+1)}{b^2}$ or substitution to solve the integral

0.5 pt: correctly stating that $\lim_{r \rightarrow +\infty} e^{-r/a} \left(\frac{r}{a} + 1\right) = 0$

1 pt: correct expression of A : $A = \frac{qe}{8\pi a^2}$

b) (9 points)

1 pt: correctly identifying and stating Gauss's Law

1 pt: correctly having that : $\oint \vec{E} \cdot d\vec{A} = 4\pi r^2$

2 pt: correctly identifying that : $Q_{\text{enc}} = q + \int_a^r \epsilon n(x) 4\pi x^2 dx$

2 pt: correctly using the formula $\int x e^{-bx} dx = -\frac{e^{-bx}(bx+1)}{b^2}$ or substitution to solve the integral

1 pt: correct final solution for E : $E(r) = \frac{qe}{8\pi \epsilon_0 r^2} e^{-r/a} \left(\frac{r}{a} + 1\right)$

0.5 pt: partially correct final solution for E (minor mistakes)

1.5 pt correct justification for the direction of the electric field

0.5 pt correct direction of the electric field : outwards

c) (5 points)

1 pt: correct formula to calculate the potential

-0.5 pt: forgetting to explicitly mention the use of $V(\infty) = 0$

2 pt: correct use of the formula $\int \left(\frac{1}{x^2} + \frac{b}{x}\right) e^{-bx} dx = -\frac{e^{-bx}}{x}$ with $b = \frac{1}{a}$

1.5 pt: correct final expression

d) (4 points)

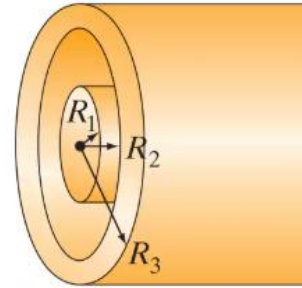
1.5 pt: correct direction of electric field

1.5 pt correct representation of equipotential lines

1 pt qualitative spacing of electric field lines

Problem 3 - 25 points

A cylindrical core of radius R_1 is made of a non-conducting material carrying a uniform and positive volumetric charge density ρ_0 . Surrounding the solid cylinder is a coaxial cylindrical shell made of a neutral conducting material which extends from inner radius R_2 to outer radius R_3 . You may assume that the length L of the cylinder and shell is such that $L \gg R_3 > R_2 > R_1$.



- a- Determine the electric charge per unit length of the cylindrical core.

Need to find charge of a length l of the core:

Volume: $V = \pi R_1^2 \cdot l$ (cylinder of radius R_1 , length l)

Charge: $Q = \rho_0 \cdot V = \rho_0 \cdot \pi R_1^2 \cdot l$

Then, the charge per length λ_1 of the core is

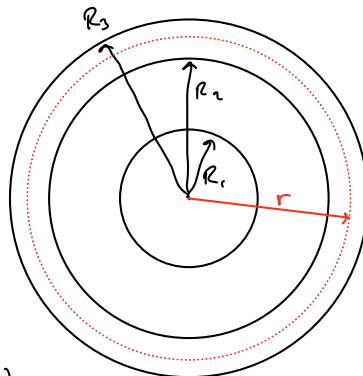
$$\lambda_1 = \frac{Q}{l} = \frac{\rho_0 \pi R_1^2 l}{l} = \boxed{\rho_0 \pi R_1^2}$$

Rubric: +5 points total

- +5: Completely correct
- +4: Minor sign/arithmetic error
- +3: Geometry error
- +2: Major conceptual error
- +1: Minimal correct work

- b- Determine the surface and volume charge densities - σ_{in} , σ_{out} and ρ_c - carried by the conducting cylindrical shell on its inner surface, outer surface and in volume, respectively. Make sure you articulate your reasoning very clearly.

Cross-sectional view:



At the radius r shown, we are inside conducting material, so $E=0$ here. Considering the dotted red line at radius r as a Gaussian cylinder of radius r and length l , the flux through

this Gaussian surface must be zero since $E=0$ on the label, and since E will be perpendicular to the top and bottom surfaces of the cylinder where E is non-zero. So, Gauss's law $\Phi_E = \frac{Q_{enc}}{\epsilon_0}$ tells us that $Q_{enc}=0$. We know the Gaussian cylinder of length l , which will fully enclose a length l of the cylindrical core, will therefore enclose a charge $Q_1 = \lambda_1 l = \rho_c \pi R_1^2 l$ from the core. To cancel this, there must be a charge $Q_{in} = -\rho_c \pi R_1^2 l$ on the conducting cylinder inside of radius r . This holds true for any radius r with $R_2 < r < R_3$, so this charge must be at the inner surface. A length l of this surface is enclosed by the Gaussian cylinder, and this corresponds to an area $A_{in} = 2\pi R_2 l$. Thus,

$$\sigma_{in} = \frac{Q_{in}}{A_{in}} = - \frac{\rho_c \pi R_1^2 l}{2\pi R_2 l} = - \frac{\rho_c R_1^2}{2R_2}$$

Changing r from R_2 to R_3 cannot change the charge enclosed, so there cannot be any volume charge density, i.e. $\rho_c = 0$. Since the conductor is net neutral, a length l of the outer surface must have charge $Q_{out} = -Q_{in} = +\rho_c \pi R_1^2 l$. This

corresponds to an area $A_{out} = 2\pi R_3 l$, so

$$\sigma_{out} = \frac{Q_{out}}{A_{out}} = \frac{\rho_c \pi R_1^2 l}{2\pi R_3 l} = \frac{\rho_c R_1^2}{2R_3}$$

In total then,

$$\sigma_{in} = - \frac{\rho_c R_1^2}{2R_2}, \quad \sigma_{out} = + \frac{\rho_c R_1^2}{2R_3}, \quad \rho_c = 0$$

Rubric: +7 points

+7: Completely correct

+6: Minor arithmetic error

+5: Sign/Geometry error

+3: Major conceptual error

+1: Minimal correct work

- c- Determine the direction and magnitude of the electric field at any radial distance from the symmetry axis. Consider $r < R_1$, $R_1 < r < R_2$, $R_2 < r < R_3$, and $r > R_3$.

All Gaussian surfaces shown are cylinders of length l , and will have flux $\Phi_E = E \cdot 2\pi r l$ through them by symmetry.

$$r < R_1: Q_{enc} = \rho_0 \cdot \pi r^2 l$$

$$\text{Gauss's law: } \Phi_E = \frac{Q_{enc}}{\epsilon_0}$$

$$\rightarrow E \cdot 2\pi r l = \rho_0 \cdot \frac{\pi r^2 l}{\epsilon_0}$$

$$\rightarrow E = \frac{\rho_0 r}{2\epsilon_0}, \text{ radially out since } \rho_0 > 0$$

$$R_1 < r < R_2: Q_{enc} = \rho_0 \cdot \pi R_1^2 l$$

$$\rightarrow E \cdot 2\pi r l = \rho_0 \cdot \pi R_1^2 l / \epsilon_0$$

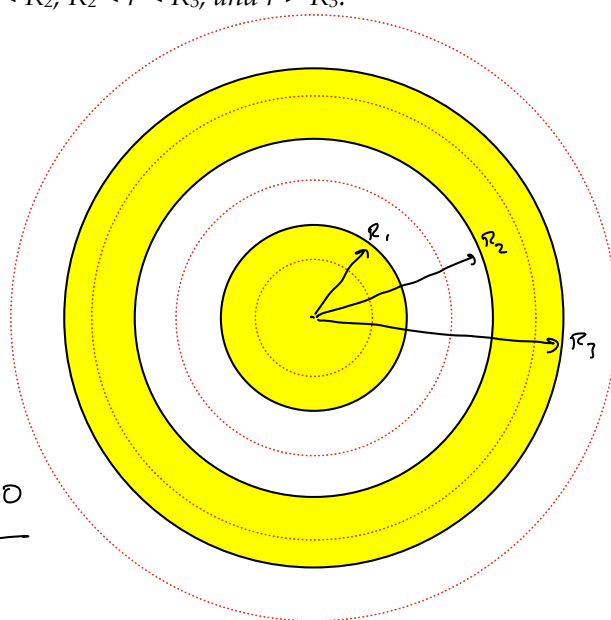
$$\rightarrow E = \frac{\rho_0 R_1^2}{2\epsilon_0 r}, \text{ radially out since } \rho_0 > 0$$

$$R_2 < r < R_3: E = 0 \text{ since inside conducting material}$$

$$r > R_3: Q_{enc} = \rho_0 \cdot \pi R_1^2 l \quad (\text{only from core since conductor is net neutral})$$

$$\rightarrow E \cdot 2\pi r l = \rho_0 \cdot \pi R_1^2 l / \epsilon_0$$

$$\rightarrow E = \frac{\rho_0 R_1^2}{2\epsilon_0 r}, \text{ radially outward since } \rho_0 > 0$$



Rubric: +10 points

Magnitudes: +7 points

+7: Completely correct

+6: Minor arithmetic error

+5: Sign/Geometry error

+3: Major conceptual error

+1: Minimal correct work

Direction: +3 points

+7: Completely correct

+2: Minor error

+1: Major errors

Name:

- d- Explain why the cylindrical core and cylindrical shell described above do not form a cylindrical capacitor.

This is not a capacitor because there is only one conductor in the system. What would be the inner plate (the cylindrical core) is insulating, and we cannot treat the inner and outer surfaces of the conductor as separate plates because they are part of the same conductor, so $V_{in} = V_{out}$ always.

Rubric: +3 points

+3: Completely correct

+2: Mostly correct with minor incorrect notions

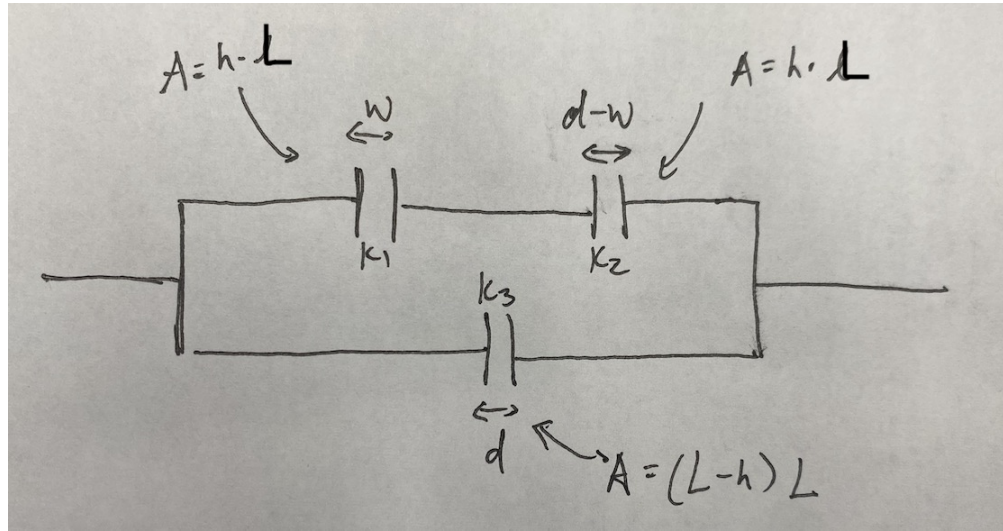
+1: Mostly incorrect with some correct notions

Physics 7B Midterm2 Sec001 Q4 Solution

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April 2, 2025

- (a) The equivalent diagram should consist of three capacitors. Two capacitors connected in series, then in parallel with a third capacitor. The dimensions are labeled in the graph below:



Since we are given the cross-section view, so the separation 'd' in the parallel-plate capacitor equation is given by d and w . The area of the capacitor is given by L and h .

- (b) First, we compute the capacitance for each capacitor by plugging in the dimensions from part(a) into the equation $C = K\epsilon_0 \frac{A}{d}$

$$\text{where: } \begin{cases} C_1 = \frac{K_1 \epsilon_0 h L}{w} \\ C_2 = \frac{K_2 \epsilon_0 h L}{d - w} \\ C_3 = \frac{K_3 \epsilon_0 (L - h) L}{d} \end{cases}$$

$$C_{\text{total}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} + C_3$$

Capacitors C_1 and C_2 are in series, so their equivalent capacitance is:

$$C_{12} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{w}{K_1 \epsilon_0 h L} + \frac{d - w}{K_2 \epsilon_0 h L} \right)^{-1} = \left(\frac{1}{\epsilon_0 h L} \left(\frac{w}{K_1} + \frac{d - w}{K_2} \right) \right)^{-1}$$

$$C_{12} = \frac{\varepsilon_0 h L}{\frac{w}{K_1} + \frac{d-w}{K_2}}$$

The total capacitance is the sum of C_{12} and C_3 , since they are in parallel:

$$C_{\text{total}} = C_{12} + C_3 = \frac{\varepsilon_0 h L}{\frac{w}{K_1} + \frac{d-w}{K_2}} + \frac{K_3 \varepsilon_0 (L - h) L}{d}$$

(c) When the capacitor is filled with material, $K > 1$. It would have several advantages:

- Increase capacitance.
- Greater energy storage.

Anything that's a result of increased capacitance should received points

(d) When a dielectric is inserted into a charged capacitor that has been disconnected from the circuit, the total charge Q on the plates remains constant. Mathematically when we think about what will happen to the Voltage difference, we have

$$V = \frac{Q}{C}$$

When Q is constant, but C increases therefore voltage will drop.

At the microscopic level, the electric field between the plates **polarizes the molecules of the dielectric**. These dipoles align with the field and create an internal electric field $\vec{E}_{\text{induced}}$ that opposes the original field $\vec{E}_0$.

Since voltage is related to electric field by

$$V = E \cdot d,$$

the new voltage becomes smaller than before because we have got a weaker field.

Therefore, inserting a dielectric reduces the voltage across the plates because the opposing internal field weakens the total field, even though the charge stays the same.

Here's a simple drawing showing what's happening microscopically:

